

$$\int_a^b f(x) dg(x)$$

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$

Chapter 6: The Riemann–Stieltjes Integral

Section: Integration and Differentiation

Based on Walter Rudin's *Principles of Mathematical Analysis*

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Outline

$$\int_a^b f(x) dg(x)$$

$$S(x) = \sum_{i=1}^n f(x_i^*) (g(x_i) - g(x_{i-1})) = g(b) - g(a)$$

Motivation

The Integral as a Function of Its Upper Limit

The Fundamental Theorem of Calculus

Integration by Parts

Summary



$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

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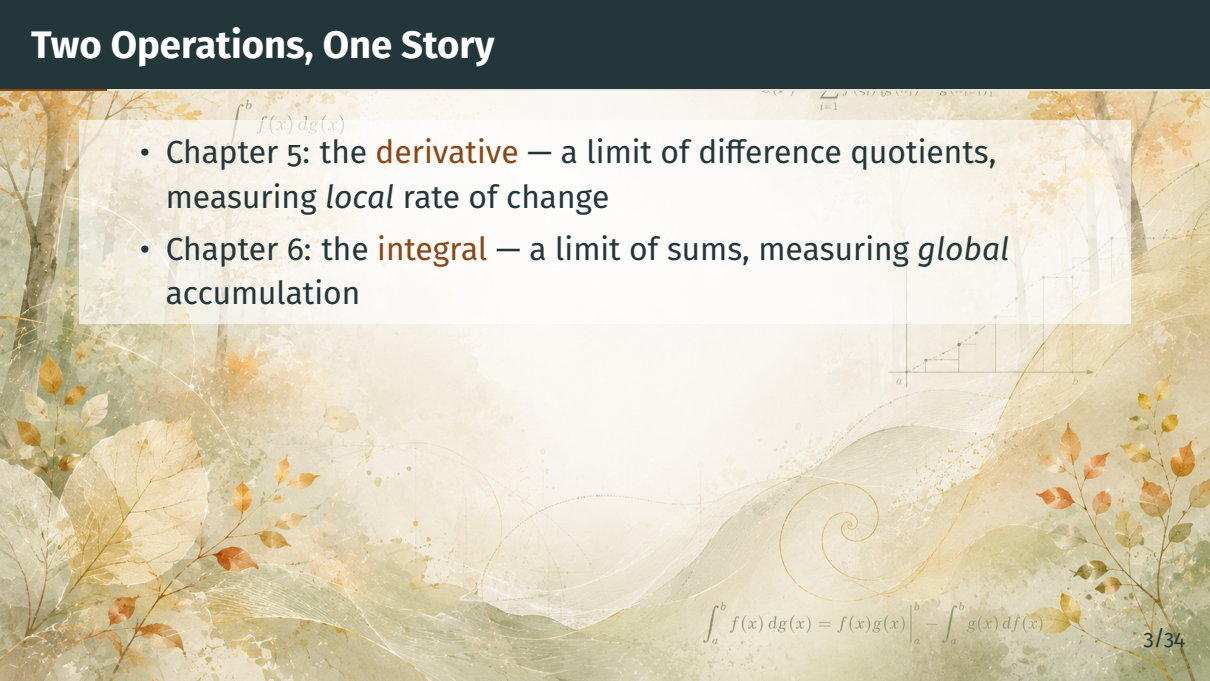
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$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Two Operations, One Story

- 
- The background features a soft-focus illustration of a forest with trees and autumn leaves in shades of orange, yellow, and green. Overlaid on this is a semi-transparent white box containing text. To the right of the text box, a small graph shows a curve with a step function approximation, with axes labeled 'a' and 'b'. At the top right, a small formula is visible: $\sum_{i=1}^n f(x_i) (g(x_i) - g(x_{i-1})) = g(b) - g(a)$.
- Chapter 5: the **derivative** — a limit of difference quotients, measuring *local* rate of change
 - Chapter 6: the **integral** — a limit of sums, measuring *global* accumulation

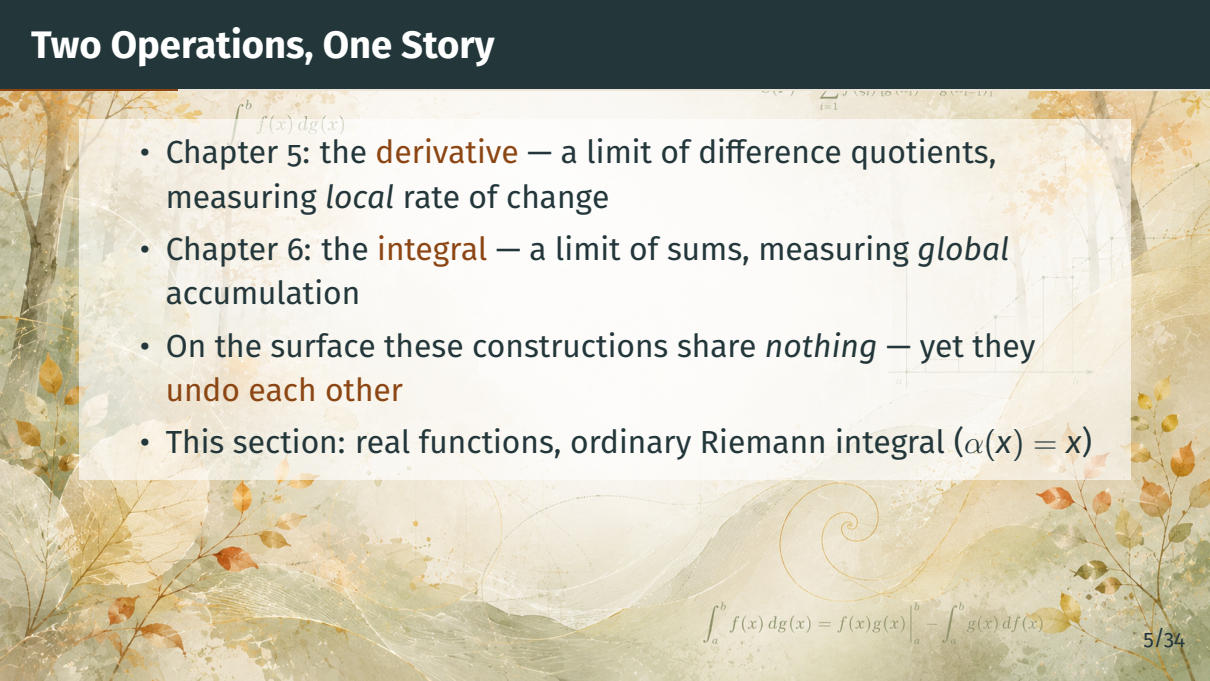
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Two Operations, One Story

- Chapter 5: the **derivative** — a limit of difference quotients, measuring *local* rate of change
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- On the surface these constructions share *nothing* — yet they **undo each other**

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 - On the surface these constructions share *nothing* — yet they **undo each other**
 - This section: real functions, ordinary Riemann integral ($\alpha(x) = x$)
- $\sum_{i=1}^n f(x_i^*) (g(x_i) - g(x_{i-1}))) = g(b) - g(a)$
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What We Will Prove

- **Theorem 6.20:** differentiate the integral $F(x) = \int_a^x f(t) dt$
 $\Rightarrow$ get f back (where f is continuous)



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The Integral as a Function of Its Upper Limit

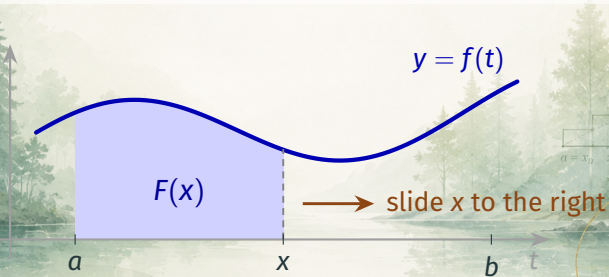


A New Function: The Area So Far

For $f \in \mathcal{R}$ on $[a, b]$ and $a \leq x \leq b$, define

$$F(x) = \int_a^x f(t) dt.$$

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$



Theorem 6.20: Differentiating the Integral

Theorem 6.20

Let $f \in \mathcal{R}$ on $[a, b]$. For $a \leq x \leq b$, put

$$F(x) = \int_a^x f(t) dt.$$

Then F is **continuous** on $[a, b]$.

Theorem 6.20: Differentiating the Integral

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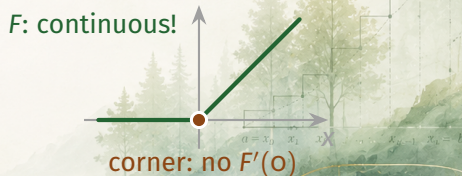
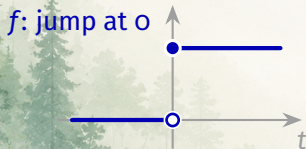
Then F is **continuous** on $[a, b]$; furthermore, if f is continuous at a point x_0 of $[a, b]$, then F is **differentiable** at x_0 , and

$$F'(x_0) = f(x_0).$$

Example: A Jump Becomes a Corner

Integrating a step

On $[-1, 1]$, let $f(t) = 0$ for $t < 0$ and $f(t) = 1$ for $t \geq 0$. Then $F(x) = \int_{-1}^x f(t) dt = \max(x, 0)$.



Proof Strategy: Theorem 6.20

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$

Two parts, two tools:

1. **Continuity of F :** the size bound of Theorem 6.12(c),(d) gives
 $|F(y) - F(x)| \leq M(y - x)$
2. **Differentiability at x_0 :** the difference quotient of F is an **average** of f , and averages of f near x_0 are close to $f(x_0)$

Recall: Theorem 6.12(c), (d)

The integral splits at any intermediate point, and $|\int f dx| \leq M \cdot$
(length of the interval) when $|f| \leq M$.

Proof of Theorem 6.20 — Part 1: Continuity

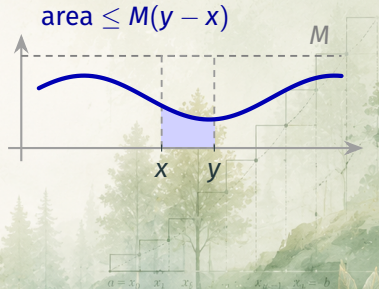
Step 1: A Lipschitz bound

$f \in \mathcal{R}$ is bounded: say $|f(t)| \leq M$. For $a \leq x < y \leq b$,

$$|F(y) - F(x)| = \left| \int_x^y f(t) dt \right| \leq M(y - x)$$

by Theorem 6.12(c) and (d).

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$



Proof of Theorem 6.20 — Part 1: Continuity

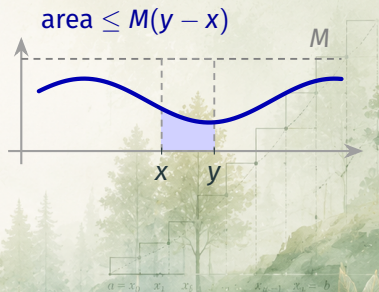
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$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$



Step 2: Uniform continuity

$$|y - x| < \varepsilon/M \implies |F(y) - F(x)| < \varepsilon.$$

F is uniformly continuous.

Proof of Theorem 6.20 — Part 2: Differentiability

Step 1: Use continuity of f at x_0

Given $\varepsilon > 0$, choose $\delta > 0$ so that $|f(u) - f(x_0)| < \varepsilon$ whenever $|u - x_0| < \delta$ and $u \in [a, b]$.

Proof of Theorem 6.20 — Part 2: Differentiability

Step 1: Use continuity of f at x_0

Given $\varepsilon > 0$, choose $\delta > 0$ so that $|f(u) - f(x_0)| < \varepsilon$ whenever $|u - x_0| < \delta$ and $u \in [a, b]$.

Step 2: The difference quotient is an average

For $x_0 - \delta < s \leq x_0 \leq t < x_0 + \delta$ (inside $[a, b]$), Theorem 6.12(d) gives:

$$\left| \frac{F(t) - F(s)}{t - s} - f(x_0) \right| = \left| \frac{1}{t - s} \int_s^t [f(u) - f(x_0)] du \right| < \varepsilon.$$

Proof of Theorem 6.20 — Part 2: Differentiability

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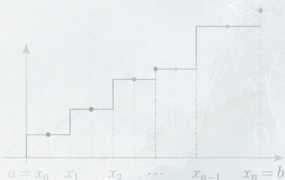
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Conclusion: letting $s, t \rightarrow x_0$: $F'(x_0) = f(x_0)$. ■

$$\int_a^b f(x) dg(x)$$

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$



The Fundamental Theorem of Calculus

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Theorem 6.21: The Fundamental Theorem of Calculus

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$

Theorem 6.21 (Fundamental Theorem of Calculus)

If $f \in \mathcal{R}$ on $[a, b]$ and if there is a differentiable function F on $[a, b]$ such that $F' = f$, then

$$\int_a^b f(x) dx = F(b) - F(a).$$

$a = x_0 \quad x_1 \quad x_2 \quad \dots \quad x_{n-1} \quad x_n = b$

In words: to integrate f , find *any* antiderivative F and take the **net change** $F(b) - F(a)$.

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Remark: Reading the Hypotheses Carefully

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$

Both hypotheses are needed

- f need *not* be continuous — integrability suffices. But integrability must be *assumed*: a derivative F' can fail to be in $\mathcal{R}$.
- If f is continuous on $[a, b]$, Theorem 6.20 provides an antiderivative automatically — both hypotheses hold for free.



$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Proof Strategy: Theorem 6.21

$$S(P) = \sum_{i=1} f(\xi_i) [g(x_i) - g(x_{i-1})]$$

Goal: $\int_a^b f dx = F(b) - F(a).$

Approach:

1. Pick a partition P with $U(P, f) - L(P, f) < \varepsilon$ (Theorem 6.6)
2. Apply the **mean value theorem** on each subinterval: the change of F across $[x_{i-1}, x_i]$ is $f(t_i) \Delta x_i$
3. Sum: the changes **telescope** to $F(b) - F(a)$: exact Riemann sum!
4. Compare this sum with the integral (Theorem 6.7(c))



$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

The Key Picture: MVT on Each Subinterval

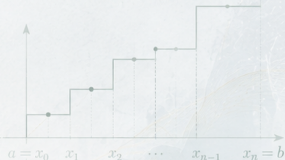
$$\int_a^b f(x) dg(x)$$

slope of chord = slope at t_2 :

$$F(x_2) - F(x_1) = f(t_2) \Delta x_2$$

$$y = F(x)$$

$$S(F) = \sum_{i=1} f(\xi_i) [g(x_i) - g(x_{i-1})]$$



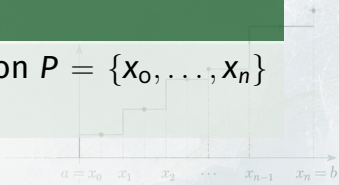
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Proof of Theorem 6.21

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$

Step 1: A fine partition

Let $\varepsilon > 0$. Since $f \in \mathcal{R}$, Theorem 6.6 gives a partition $P = \{x_0, \dots, x_n\}$ with $U(P, f) - L(P, f) < \varepsilon$.


$$a = x_0 \quad x_1 \quad x_2 \quad \dots \quad x_{n-1} \quad x_n = b$$

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

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Step 2: Mean value theorem on each subinterval

Since $F' = f$, the MVT (Theorem 5.10) furnishes points $t_i \in [x_{i-1}, x_i]$, $i = 1, \dots, n$, with $F(x_i) - F(x_{i-1}) = f(t_i) \Delta x_i$.

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

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Step 3: The sum telescopes

$$\sum_{i=1}^n f(t_i) \Delta x_i = \sum_{i=1}^n [F(x_i) - F(x_{i-1})] = \boxed{F(b) - F(a)}.$$

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Proof of Theorem 6.21

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$

Steps 1–3 (summary)

$U(P, f) - L(P, f) < \varepsilon$, and the MVT gives $\sum_{i=1}^n f(t_i) \Delta x_i = F(b) - F(a)$.

Step 4: Compare with the integral

Theorem 6.7(c): every Riemann sum on such a P is within ε of the integral:

$$\left| F(b) - F(a) - \int_a^b f(x) dx \right| < \varepsilon.$$

True for every $\varepsilon > 0$, so

$$\int_a^b f(x) dx = F(b) - F(a).$$



$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

$$\int_a^b f(x) dg(x)$$

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$

Integration by Parts



$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Theorem 6.22: Integration by Parts

Theorem 6.22 (integration by parts)

Suppose F and G are differentiable functions on $[a, b]$, $F' = f \in \mathcal{R}$, and $G' = g \in \mathcal{R}$. Then

$$\int_a^b F(x)g(x) dx = F(b)G(b) - F(a)G(a) - \int_a^b f(x)G(x) dx.$$

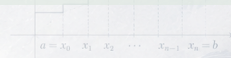
Key idea: the **product rule** $(FG)' = Fg + fG$, run through the FTC.

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Proof of Theorem 6.22

Step 1: Set up the product

Put $H(x) = F(x)G(x)$. By the product rule (Theorem 5.3), $H' = Fg + fG$.
 $H' \in \mathcal{R}$: F, G are continuous, so $Fg, fG \in \mathcal{R}$ by Theorems 6.11 and 6.13.



$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

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Step 2: Apply the FTC to H

Theorem 6.21: $\int_a^b [Fg + fG] dx = H(b) - H(a)$. Splitting (linearity) and rearranging:

$$\int_a^b Fg dx = F(b)G(b) - F(a)G(a) - \int_a^b fG dx. \quad \blacksquare$$

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Remark: A Discrete Cousin — Summation by Parts

$$\int_a^b f(x) dg(x)$$

Recall: Theorem 3.41 (partial summation formula)

With $A_n = \sum_{k=0}^n a_k$ (and $A_{-1} = 0$), for $0 \leq p \leq q$:

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$

$$\sum_{n=p}^q a_n b_n = A_q b_q - A_{p-1} b_p - \sum_{n=p}^{q-1} A_n (b_{n+1} - b_n).$$



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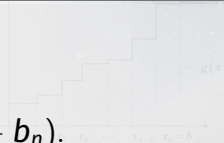
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$$\sum_{n=p}^q a_n b_n = A_q b_q - A_{p-1} b_p - \sum_{n=p}^{q-1} A_n (b_{n+1} - b_n).$$



Same identity as Theorem 6.22: $\int \leftrightarrow \sum$, derivative $\leftrightarrow$ **difference**:

$$G \leftrightarrow A_n, \quad g = G' \leftrightarrow a_n = A_n - A_{n-1}, \quad F \leftrightarrow b_n, \quad f = F' \leftrightarrow b_{n+1} - b_n.$$

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

$$\int_a^b f(x) dg(x)$$

Summary



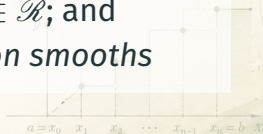
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Summary

Key takeaways from this section:

- **Theorem 6.20:** $F(x) = \int_a^x f(t) dt$ is **continuous** for $f \in \mathcal{R}$; and $F'(x_0) = f(x_0)$ wherever f is continuous — *integration smooths*

$$\int_a^x f(x) dg(x)$$



$$\sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$

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$$\int_a^x f(x) dg(x)$$

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- **Theorem 6.20:** $F(x) = \int_a^x f(t) dt$ is **continuous** for $f \in \mathcal{R}$; and $F'(x_0) = f(x_0)$ wherever f is continuous — *integration smooths*
- **Theorem 6.21 (FTC):** $f \in \mathcal{R}$ and $F' = f \implies \int_a^b f dx = F(b) - F(a)$ (proof: MVT + telescoping + Theorem 6.7(c))

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- **Theorem 6.20:** $F(x) = \int_a^x f(t) dt$ is **continuous** for $f \in \mathcal{R}$; and $F'(x_0) = f(x_0)$ wherever f is continuous — *integration smooths*
- **Theorem 6.21 (FTC):** $f \in \mathcal{R}$ and $F' = f \implies \int_a^b f dx = F(b) - F(a)$ (proof: MVT + telescoping + Theorem 6.7(c))
- **Theorem 6.22:** integration by parts — the product rule through the FTC: $\int Fg = FG \Big|_a^b - \int fG$

Coming up next: integration of **vector-valued** functions, and the length of a curve.